

REAL ANALYSIS 1 MIDTERM EXAM

This exam is worth a total of 50 points.

Problem 1 (10 points)

Let X be a countable set and $A \subseteq X$. Prove that A is countable.

Solution.

If X is finite in cardinality there is nothing to show since $A \subseteq X$ means that A must then be finite. Similarly, if X is countably infinite but A is finite we are done. Assume both A, X are infinite in cardinality. Since X is countable there is a bijection:

$$\varphi : X \rightarrow \mathbb{N}$$

implying that the restriction $\varphi|_A : A \rightarrow \mathbb{N}$ is an injection.

Problem 2 (10 points)

Prove by induction that

$$\underbrace{\sqrt{2 + \sqrt{2 + \sqrt{2 \cdots + \sqrt{2}}}}}_{n \text{ nested square roots}} = 2 \cos \left(\frac{\pi}{2^{n+1}} \right)$$

for all $n \in \mathbb{N}$.

Solution.

Consider $n = 1$, we have $2 \cos \left(\frac{\pi}{2^{n+1}} \right) = 2 \cos \left(\frac{\pi}{4} \right) = 2 \cdot \frac{1}{\sqrt{2}} = \sqrt{2}$, which establishes the base-case. Now assume the equation above holds up to $n \in \mathbb{N}$, we show the case for $n + 1$. Consider now:

$$\underbrace{\sqrt{2 + \sqrt{2 + \sqrt{2 \cdots + \sqrt{2}}}}}_{n+1 \text{ nested square roots}} \stackrel{\text{ind}}{=} \sqrt{2 + 2 \cos \left(\frac{\pi}{2^{n+1}} \right)} \quad (\star)$$

We now recall the half-angle formula:

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}, \quad \forall \theta \in \mathbb{R}$$

The equation in $(\star)$ now becomes:

$$\begin{aligned} \sqrt{4 \left(\frac{1 + \cos\left(\frac{\pi}{2^{n+1}}\right)}{2} \right)} &= \sqrt{4 \cos^2\left(\frac{\pi}{2^{n+2}}\right)} \\ &= 2 \cos\left(\frac{\pi}{2^{n+2}}\right) \end{aligned}$$

as was required.

Problem 3 (10 points)

Prove the following using only the $\varepsilon - N$ definition of the limit:

(a) (5 points) $\lim_{n \rightarrow \infty} \frac{n^2 - 1}{2n^2 + 3} = \frac{1}{2}$

(b) (5 points) $\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n^2 + 1} = 0$.

Solution.

(a) We first estimate the difference:

$$\begin{aligned} \left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| &= \left| \frac{2n^2 - 2 - 2n^2 - 3}{2(2n^2 + 3)} \right| = \frac{5}{2|2n^2 + 3|} \\ &< \frac{5}{|2n^2 + 3|} \\ &< \frac{5}{2n^2} \end{aligned}$$

Fix $\varepsilon > 0$, and observe that $\frac{1}{n^2} \rightarrow 0$ as $n \rightarrow \infty$. Then, there is some $N \in \mathbb{N}$ such that

$$\frac{1}{n^2} < \frac{2\varepsilon}{5}, \quad \forall n \geq N$$

Now, for all such n we note that:

$$\left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| < \frac{5}{2n^2} < \frac{5}{2} \cdot \frac{2\varepsilon}{5} = \varepsilon$$

(b) The procedure here is similar, in absolute value we find for all $n \in \mathbb{N}$:

$$\left| \frac{\sqrt{n}}{n^2 + 1} \right| < \frac{\sqrt{n}}{n^2} \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}$$

Now, recalling that

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$$

for each $\varepsilon > 0$ we may choose $N \in \mathbb{N}$ such that $\frac{1}{\sqrt{n}} < \varepsilon$ for all $n \geq N$. Therefore, we indeed have established (b).

Problem 4 (10 points)

Let $f, g : [a, b] \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and assume that f is bounded. If for some $x_0 \in (a, b)$

$$\lim_{x \rightarrow x_0} g(x) = 0$$

prove that

$$\lim_{x \rightarrow x_0} f(x)g(x) = 0$$

Solution.

By hypothesis there exists some $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a, b]$. Let $\varepsilon > 0$ be given, since

$$\lim_{x \rightarrow x_0} g(x) = 0$$

there is some $\delta > 0$ such that $|g(x)| < \frac{\varepsilon}{M}$ for all $x \in [a, b]$ with $0 < |x - x_0| < \delta$. Now, for all such values of x we have the following:

$$|f(x)g(x) - 0| = |f(x)g(x)| = |f(x)| |g(x)| \leq M |g(x)| < M \cdot \frac{\varepsilon}{M}$$

implying that $|f(x)g(x)| < \varepsilon$ for all $x \in V_\delta^*(x_0) \cap [a, b]$. Since $\varepsilon > 0$ was taken arbitrarily this implies that

$$\lim_{x \rightarrow x_0} f(x)g(x) = 0$$

as was asserted.

Problem 5 (10 points)

(a) (5 points) Prove the reverse triangle inequality:

$$||a| - |b|| \leq |a - b|, \quad \forall a, b \in \mathbb{R}$$

(b) (5 points) Using the fact that $\mathbb{Q}$ is a dense subspace of $\mathbb{R}$ show that for each $x \in \mathbb{R}$ and $\varepsilon > 0$ there exist $q, r \in \mathbb{Q}$ with $q < x < r$ and $|q - r| \leq \varepsilon$.

Solution.

(a) Fix $a, b \in \mathbb{R}$. Observe first that

$$|a| = |a - b + b| \leq |a - b| + |b|$$

yielding $|a| - |b| \leq |a - b|$. Reversing the roles of a, b we find

$$|b| \leq |b - a| + |a|$$

whence $|b| - |a| \leq |a - b|$, or, rather $|a| - |b| \geq -|a - b|$. These inequalities imply:

$$-|a - b| \leq |a| - |b| \leq |a - b|$$

and thus $||a| - |b|| \leq |a - b|$.

(b) Let $x \in \mathbb{R}$ be given and fix $\varepsilon > 0$. By the density of $\mathbb{Q}$ in $\mathbb{R}$ we have

$$\mathbb{Q} \cap \left(x - \frac{\varepsilon}{2}, x\right) \neq \emptyset$$

and thus we may pick $q \in \mathbb{Q}$ with $q < x$ and $|x - q| < \frac{\varepsilon}{2}$. Similarly, we pick a rational $r \in \mathbb{Q} \cap (x, x + \frac{\varepsilon}{2})$. Then we have

$$|q - r| = |q - x + x - r| \leq |q - x| + |x - r| < 2 \cdot \frac{\varepsilon}{2} = \varepsilon$$